

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000ES206B - PROGRAMMING FOR PROBLEM SOLVING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is NOT an input device?
A. Scanner B. Plotter C. Keyboard D. Microphone
- 2 In a computer, the CPU is made up of:
A. ALU and RAM B. ALU and CU C. RAM and ROM D. CU and ROM
- 3 Which symbol is commonly used for a decision in a flowchart?
A. Rectangle B. Oval C. Parallelogram D. Diamond
- 4 What is the size of a float data type typically on a 32-bit system?
A. 2 bytes B. 4 bytes C. 6 bytes D. 8 bytes
- 5 What is the result of the expression $(5 > 3) \&\& (4 < 2)$?
A. 1 B. 0 C. True D. Error
- 6 What is the output type of this expression: $10 / 4$ in C?
A. 2.5 B. 2 C. 2.0 D. 2.50
- 7 Which control structure is best suited when there are multiple constant values to compare against a single variable?
A. if-else ladder B. switch-case C. nested if D. goto
- 8 Which of the following loops guarantees at least one execution?
A. while B. for C. do-while D. None of the above
- 9 Which keyword is used to define a function in C?
A. function B. define C. return D. Data type like int, void
- 10 Which of the following is the correct way to declare a 2D array of 3 rows and 4 columns?
A. `int arr[4][3];` B. `int arr[3][4];` C. `int arr[3][4]` D. `int arr{3,4};`
- 11 Which operator is used to access structure members through a pointer?
A. . B. * C. -> D. &
- 12 What function is used to close a file in C?
A. `closefile()` B. `fclose()` C. `fileclose()` D. `close()`

Q2. Solve the following.

- A) Write a detailed note on the types of computer software with suitable examples. 6
- B) Explain the steps involved in program development. 6

Q3. Solve the following.

- A) Describe various data types available in C along with their sizes and ranges. 6
- B) Write a C program that takes marks of 3 subjects and calculates total, average, and grade (use conditional operators and logical expressions). Grade ranges: (Grade A: $\text{Average} \geq 85$, Grade B: $70 \leq \text{Average} < 85$, Grade C: $50 \leq \text{Average} < 70$, Grade D: $\text{Average} < 50$, F: $\text{Average} < 40$) 6

Q4. Solve Any Two of the following.

- A) Define a function. How are user-defined functions declared, defined, and called in C? 6
- B) Create a menu-driven C program using functions to perform arithmetic operations (+, -, ×, ÷) repeatedly until the user chooses to exit. 6
- C) Explain the use of break, continue, and goto statements with appropriate code snippets. 6

Q5. Solve Any Two of the following.

- A) Define a pointer. Explain how pointers can be used to access array elements with examples. 6
- B) Describe multidimensional arrays with syntax and example of matrix addition. 6
- C) Write a C program that uses a character pointer to read and print a string, and count the number of vowels using pointer traversal. 6

Q6. Solve Any Two of the following.

- A) Explain how structures can be passed to functions by value and by reference with examples. 6
- B) Create a structure Book with members title, author, price. Write a program to input details of 3 books, and display books with price > ₹500. 6
- C) Following C program is to accept student details (roll number, name, marks) from the user and save them to a text file (record.txt) using file handling and structures. Find Errors in following code snippet. 6

```
#include<stdio.h>
struct Student {
    int roll;
    char name[20];
    float marks;
};
int main() {
    Student s1;
    printf("Enter details: ");
    scanf("%d %s %f", &s1.roll, s1.name, s1.marks);
    FILE fp;
    fp = fopen("record.txt", "w");
    fprintf(fp, "%d %s %.2f", s1.roll, s1.name, s1.marks);
    fclose(fp);
    return 0;
}
```

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